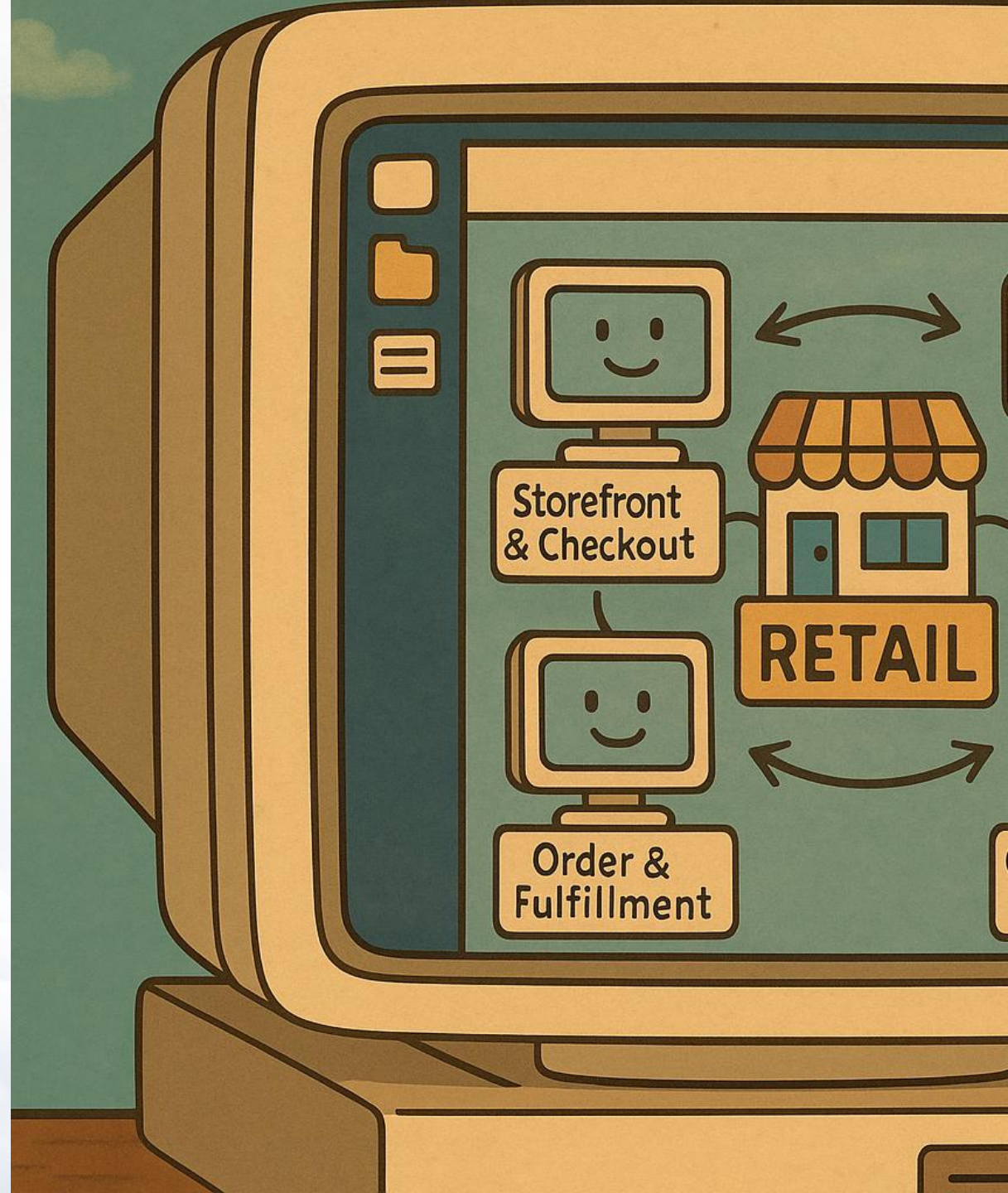




Retail and Consumer Goods: Apps that matter

Disclaimer: This content is for Microsoft partner guidance only and not to be shared directly with customers.



What is an AI frontier firm

→ The Frontier Firm Defined

A next-generation company that fuses human judgment with AI agents to scale faster, work smarter, and unlock new value - powered by on-demand intelligence and human + agent hybrid teams.

💡 Core Characteristics

- **AI-operated, human-led:** Agents execute tasks; humans provide oversight.
- **Work Chart vs Org Chart:** Teams form around outcomes, not departments.
- **Agent boss mindset:** Every employee manages and collaborates with AI agents.

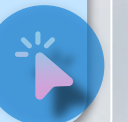
L Why This Matters Now

- **82%** of leaders say 2025 is a pivotal year to rethink strategy and operations.*
- **80%** expect agents to be integrated into their AI strategy within 12–18 months.*
- Frontier Firms outperform peers in productivity, innovation, and speed to value

"As AI democratizes access to expertise and intelligence, we'll see the rise of Intelligence Resources departments -much like how HR and IT evolved into core functions. These new divisions will be essential for managing the interplay between humans and AI agents, emerging as a critical source of competitive advantage in the AI-enabled enterprise."

-Karim R. Lakhani, Chair of the Digital, Data, and Design Institute at Harvard,

[Microsoft Work Trends Index Report](#)



AI Apps that matter:

Build AI Apps that drive Frontier Transformation

Business Impact

Delivers clear, measurable value by solving a real problem aligned with user needs and organizational goals: high-value use cases

Innovative

Leverages cutting-edge agentic technologies and real time data to solve complex problems in ways that are innovative, adaptive, and often predictive or autonomous

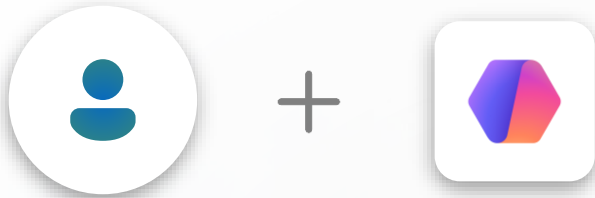
Enterprise-grade

Production-grade systems that are governed, secure, trusted, operationally stable, business-aligned, continuously optimized

Journey to the Frontier Firm

Phase 1

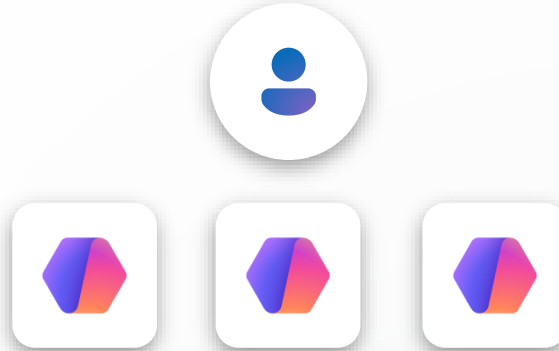
Human with assistant



Every employee has an AI assistant that helps them work better and faster

Phase 2

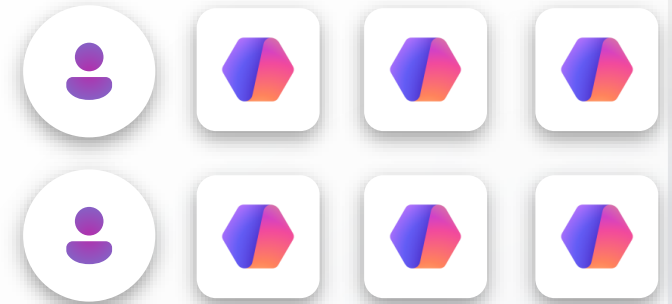
Human-led agents



Agents join teams as "digital colleagues," taking on specific tasks at human direction

Phase 3

Human-led, agent-operated



Humans set direction and agents run entire business processes and workflows, checking in as needed

Becoming Frontier

Success framework

Enrich employee experiences

Reinvent customer engagement

Reshape business processes

Bend the curve on innovation



Approach



Adopt Microsoft copilot



Build in Copilot Studio



Build on Microsoft Azure

Guardrails on how to work with the suggested processes

See the whole process



Map the end-to-end flow, actors, policies, and desired outcomes

Call out where agents could take actions

Decompose into "agent-able" jobs



Prioritize by value $\times$ feasibility $\times$ data readiness

Define success metrics and guardrails; timebox the first increment

Select the starting "slice"



Break into sub-processes with clear inputs, decisions, actions, outputs

Create a data map: sources, owners, contracts, latency, quality

Build an enterprise-ready agent



Design for extensibility: APIs/events, modular tools, config over code

Safety & compliance: identity/permissions, policy/RAG, tracing

Scale along the chain



Add/compose agents, automate handoffs, integrate with systems of record

Track ROI; harden to repeatable IP and reuse components

Ground rules: Agentic processes are a combination of multiple software solutions

Retail&CG frontier solutions

Overall industry specifics:

- Augmentation-first for brand safety & compliance: agents draft, recommend, and assist; humans approve for anything public-facing, full auditability and rollbacks
- Composable over custom: lean on commerce platforms - agents orchestrate via APIs and webhooks rather than replace systems

1	Interactive brand and product agent	Brand-owned shopping agents deliver one-to-one guidance inside the brand experience, remember preferences, advise on fit/style, answer questions, and hand off to checkout, building loyalty across channels	
2	Content gen. and operations	Turn a unified customer view into plans, assets, and delivery - accelerating campaign creation and execution with real-time and predictive insights to boost ROI and keep messages consistent across channels.	
3	Agentic commerce (shopping through 3 rd -party agent)	Orchestrate catalog, pricing and availability under clear policies; expose secure, well-documented actions and data endpoints so customers and third-party agents can discover and purchase, expanding reach	
4	Inventory planning and trend forecast	Fuse real-time signals from POS, e-commerce, weather, and promos with predictive models to anticipate demand by SKU/location, reducing stockouts and obsolescence, aligning production, and lifting service levels	
5	R&D and product design	Turn trends, signals, and regulatory shifts into portfolio decisions - surface gaps, auto-draft concept briefs and targeted design tweaks, and support gated experiments to compress time-to-market and launch success	

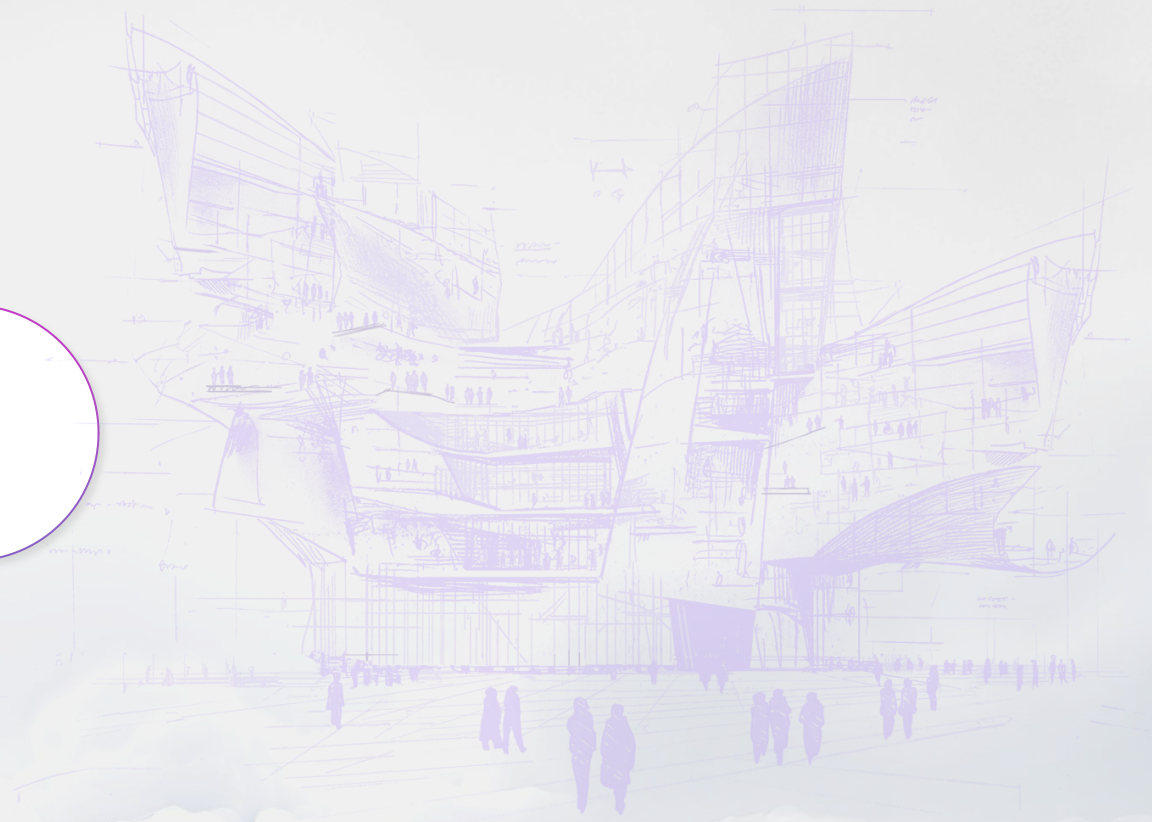
Marketing

Supply chain

Key metrics relevant to the industry

- **Growth & conversion:** conversion rate, AOV, repeat purchase/CLV
- **Content velocity & efficiency:** time-to-publish, cost per asset, engagement/CTR
- **Commerce ops & reach:** checkout rate, cart abandonment, marketplace revenue mix
- **Supply & availability:** forecast accuracy, stockouts/OTIF, inventory turns
- **Innovation speed & quality:** time-to-market, experiment cycle time, defect/COGS impact

Brand agent



1. Interactive brand and product agent

Guided shopping supports a customer from first interest to confident purchase. The process collects intent and preferences, explains products in plain language, compares options, confirms availability and delivery dates, applies prices and offers, and either completes checkout or passes the customer to a human associate with full context, while staying on brand and within legal guidelines.

What - Process overview



Current challenges

- Discovery is fragmented; filters miss real intent and advice differs by channel.
- Product pages are dense; customers repeat questions to service teams.
- Missing or inconsistent attributes reduce recommendation quality and increase returns.
- Inventory and prices change quickly; delivery promises are unclear.
- Brand, legal, and safety language is hard to control across regions and languages.



Frontier solution

Process description

- Capture intent, preferences, size, compatibility, and budget through conversation.
- Retrieve trusted facts, guides, and reviews; explain trade-offs in simple terms.
- Recommend items and bundles; compare alternatives based on needs and constraints.
- Check store and warehouse stock by location; propose substitutes or pre-orders.
- Apply promotions and coupons under guardrails; show delivery or pickup dates.
- Add items to cart or book an appointment; hand off to an associate with context.

Solution description

Agents collaborate via events, share context, and hand off to humans for final sign-off:

- **Discovery & fit agent:** Interprets intent and maps needs to product attributes, size charts, and compatibility rules
- **Catalog Q&A Agent:** Answers product questions using approved manuals, care guides, and reviews, flags gaps in content
- **Price & Offer Agent:** Applies promotions and coupons under guardrails (floors, parity, legal claims), calculates the final price
- **Availability & Promise Agent:** Checks stock by location, proposes substitutes or preorders
- **Brand & Policy Guardrail Agent:** Enforces tone of voice, safety language, and legal claims, routes edge cases to human

Why – Impact¹



Business Outcomes²

Cost savings:

- Fewer service contacts by answering product questions and guiding checkout (15–30% reduction in chat and call volumes)
- Lower return and exchange rates through better fit and compatibility guidance (5–15% fewer returns)

Growth

- Conversion & basket size: higher conversion rate and AOV via guided recs, bundles, and timely substitutes
- Loyalty & engagement: higher repeat purchase rate, active-member share, and top-of-funnel engagement from premium experiences



Roles addressed

- CMO / CDO
- Ecommerce managers and digital product owners
- Merchandising and category managers

How – Underlying foundation¹

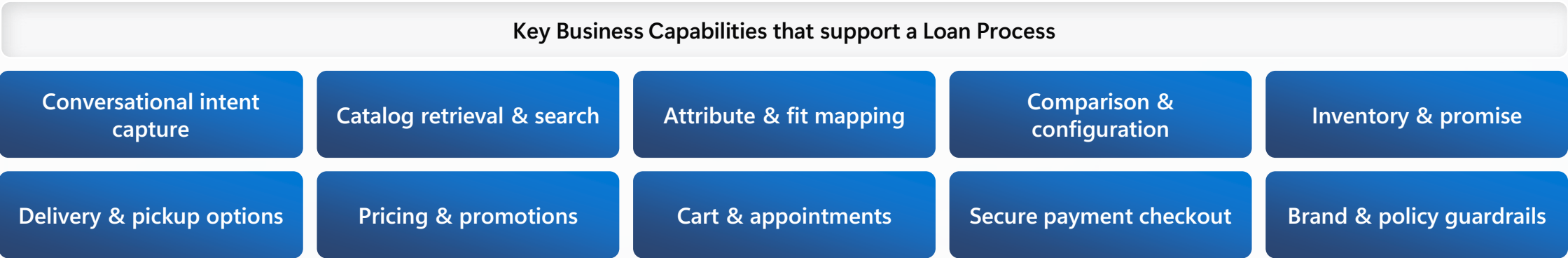
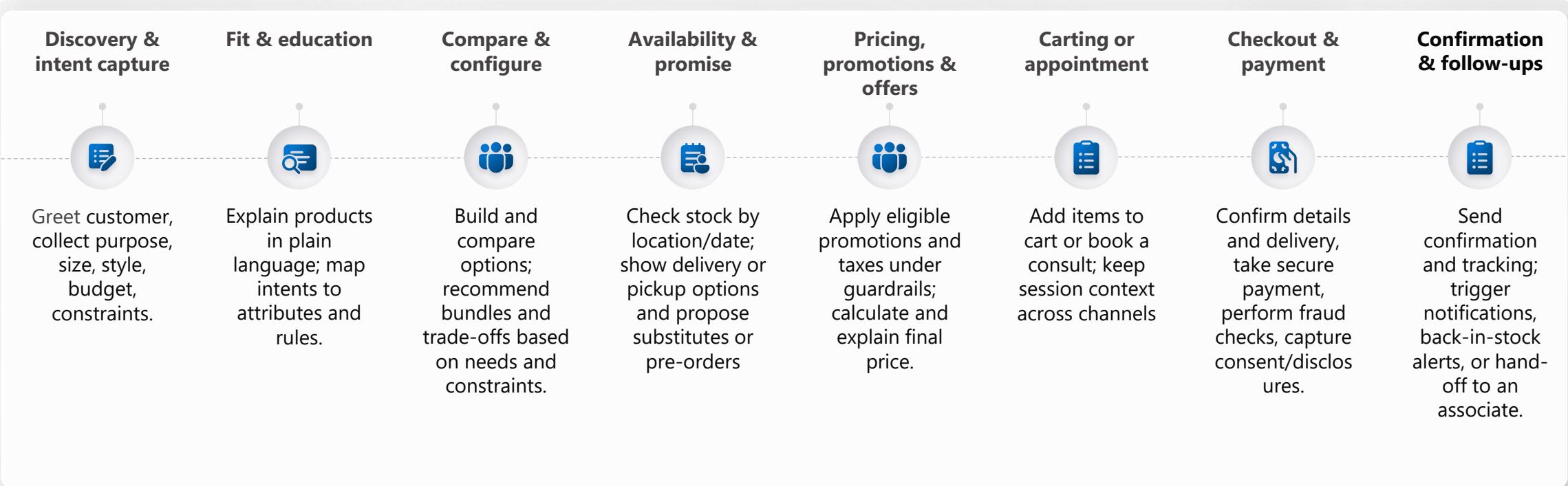
Data to integrate

- Product catalog with attributes, variants, sizes, compatibility rules, images & video
- Pricing, promotions, coupons, taxes, and legal or brand claim guidelines
- Inventory by location, delivery calendars, pickup options, and shipping service levels

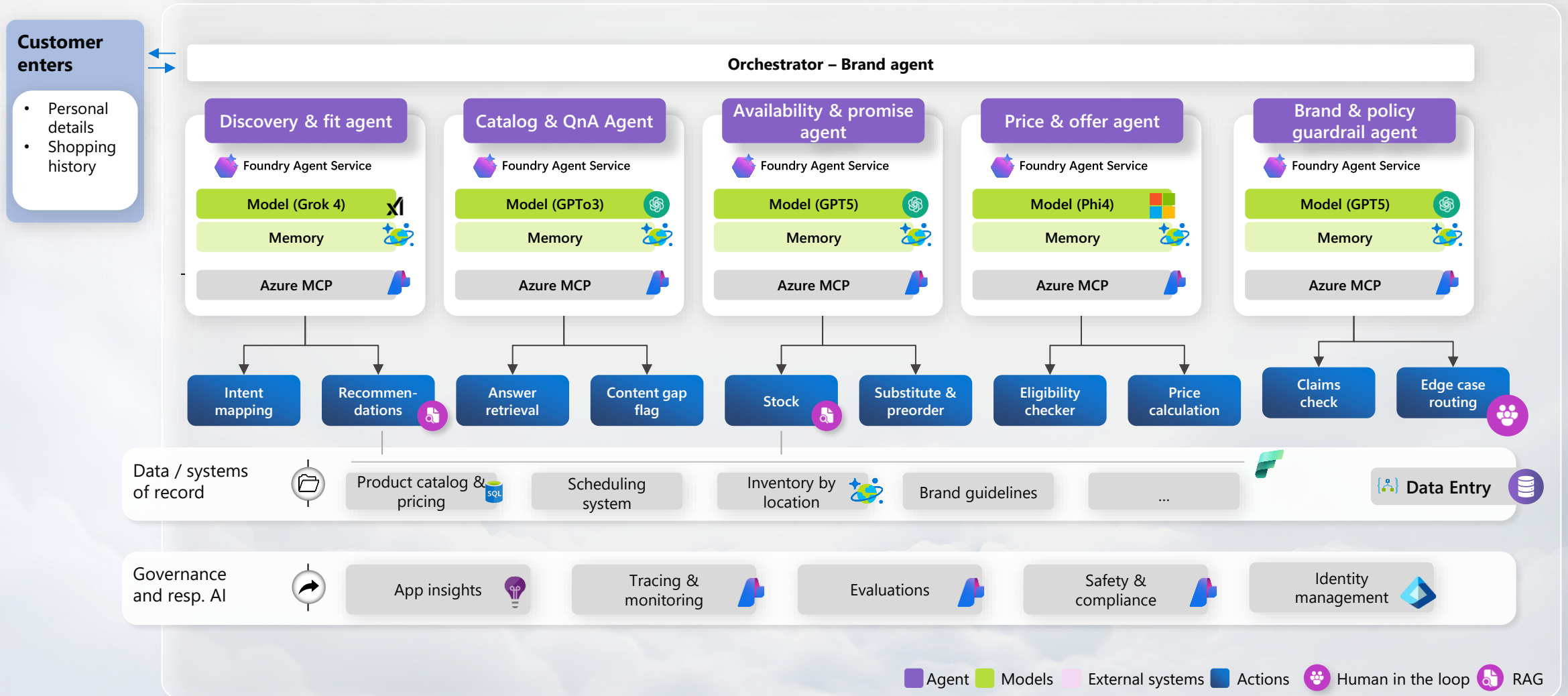
Technical patterns

- Retrieval over product catalog, policies, and guides with tool-use into the commerce system (search, price, availability, add to cart)
- Real-time personalization using behavioral signals and business rules, with clear guardrails

1. What does the shopping work process look like?



1. Example agentic shopping with brand agents





Content generation & adoption

2. Content generation and adoption

Content generation and operations turn a unified customer view into plans, assets, and delivery – accelerating campaign creation and execution with real-time and predictive insights to increase return on investment and keep messages consistent across channels.

What - Process overview



Current challenges

- Briefs, calendars, and assets live in separate tools; handoffs are slow and error-prone.
- Creative work is manual; localization and variants delay launches and raise costs.
- Brand voice and legal language are inconsistent across teams, regions, and partners.
- Performance data rarely feeds back into creative choices in time to matter.
- Compliance reviews create bottlenecks, especially for regulated products and claims.



Frontier solution

Process description

- Build an audience brief from customer segments, goals, budget, and timing.
- Plan campaigns and channel mix; generate outlines and creative concepts.
- Produce copy, images, and video variants; adapt for regions and languages.
- Apply brand and legal guardrails; capture approvals with full traceability.
- Schedule and deliver to each channel; tailor format and placements automatically.
- Monitor results and learnings; update briefs and assets for the next cycle.

Solution description

Agents collaborate via events, share context, and hand off to humans for final sign-off:

- **Audience & Brief Agent:** Translates goals and customer segments into a campaign brief, themes, calendar, and success measures.
- **Creative & Localization Agent:** Drafts copy, images, and video variants; adapts for language, region, and audience while preserving brand voice.
- **Channel Orchestration & Insights Agent:** Formats for each channel, schedules delivery, reads performance signals, and recommends next actions.
- **Brand & Policy Guardrail Agent:** Enforces tone, claims, and legal rules; manages approvals and escalates edge cases to humans.

Why – Impact¹



Business Outcomes²

Cost savings:

- 20–40% lower content production and adaptation costs (fewer revisions)
- 30–50% faster time to launch by accelerating approvals
- Savings in agency billable hours by improving creative brief process

Growth

- More relevant messages lift engagement and conversion (5–15% improvement across key channels)
- Higher ROAS (return on ad spend) due to faster content iterations



Roles addressed

- Agencies
- Marketing operations leaders and campaign managers.
- Content and creative directors.

How – Underlying foundation¹

Data to integrate

- Consented customer profiles, segments, and behavioral signals.
- Brand guidelines, prior assets, templates, and approved claims.
- Channel and campaign performance data (delivery, engagement, conversion).

Technical patterns

- Retrieval over brand rules, product information, and asset libraries with tool use into content and design systems (create, adapt, publish).
- Event-driven orchestration with real-time feedback loops to test variations, learn, and update briefs and assets under clear guardrails.

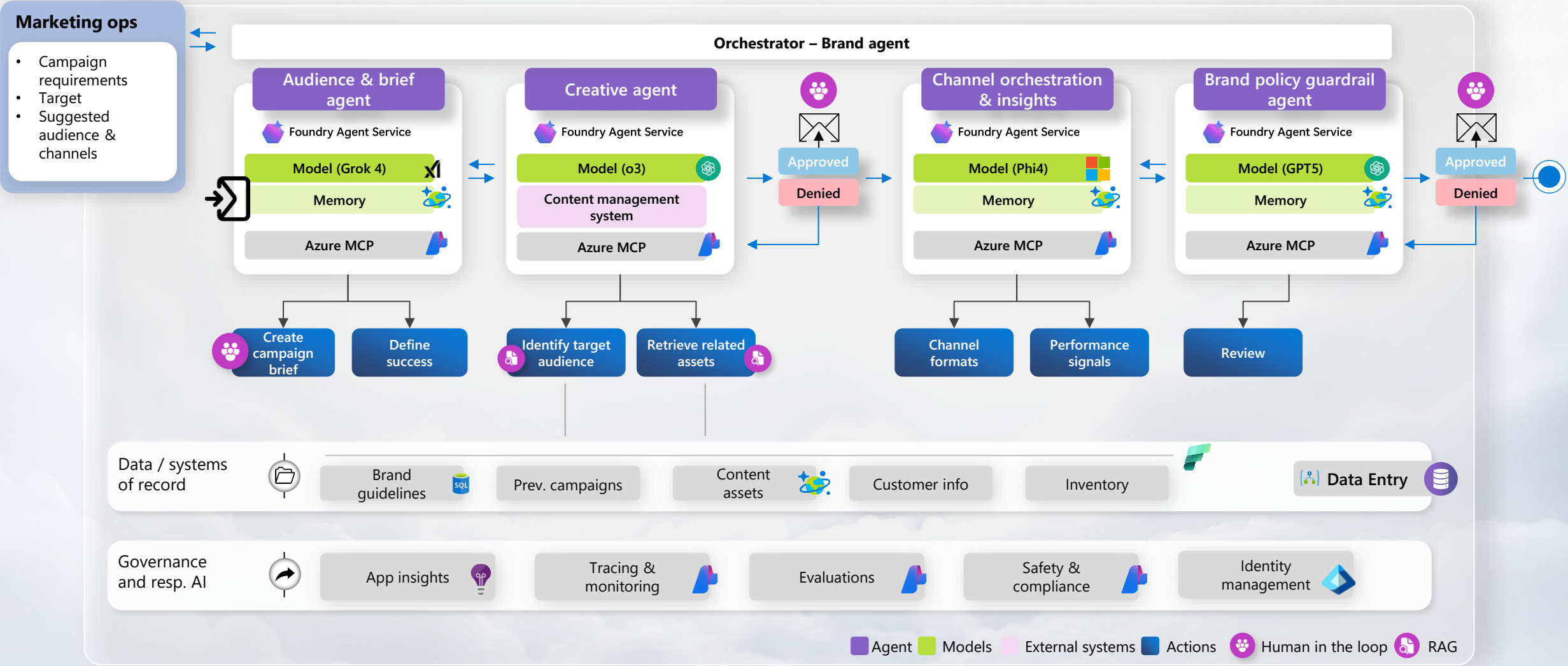
2. What does the content generation and ops process look like?



Key Business Capabilities required for this process

Audience segmentation & insights	Brief and goals capture	Content retrieval & reuse	Localization & variant management	Brand voice guardrails
Legal claims management	Approval workflow & traceability	Creative generation & editing	Channel formatting & publishing	Performance analytics & learning

2. Example agentic content generation and adoption



Agentic commerce

- shopping through 3rd-party Copilot

3. Agentic commerce

A brand-owned commerce agent orchestrates product catalog, pricing, availability, content, and fulfillment under clear policies. It exposes secure, well-documented actions and data endpoints so customers, and third-party agents, can discover, query, compare, and purchase from the brand. This expands reach across agent ecosystems while the brand retains control of experience, policy, and margins.

What - Process overview



Current challenges

- Research happens across many tabs; specifications differ by retailer and are hard to compare.
- Prices, stock, delivery promises, and promotions change quickly; signals are inconsistent across channels.
- Reviews and content are noisy; benefits and trade-offs are unclear at a glance.
- Visual discovery is limited; customers cannot easily search with photos or natural language.
- Checkout requires site switching and re-entering details, causing abandonment and support contacts.



Frontier solution

Process description

- Capture a shopper's need in plain language or through a photo; suggest comparable products.
- Normalize specifications across the brand and sellers; summarize pros and cons from trusted content.
- Track live prices, availability, delivery options, and promotions with transparent rules.
- Present best-fit choices and bundles with clear reasoning and merchant neutrality.
- Build carts and complete a protected checkout (or deep-link) with confirmations and returns guidance.
- Expose the same data and actions through documented endpoints so external agents can discover, compare, and purchase with consent.

Solution description

Agents collaborate via events, share context, and hand off to humans for final sign-off:

- **Discovery & Comparison Agent** understands a need (text, voice, or photo), finds candidates, and normalizes specifications
- **Price & Availability Agent** monitors live prices, delivery dates, stock, and reviews
- **Offer & Policy Agent** applies promotion rules and required disclosures (e.g., sponsored or affiliate), ensures legal and brand safety
- **Checkout Agent** builds cart per merchant, supports universal checkout, passes context securely, and returns order status.
- **Data & Action Exposure:** Public, policy-enforced façade that publishes machine-readable data, exposes purchase actions through A2A with authentication and consent

Why – Impact¹



Business Outcomes²

Cost savings:

- About 10–20% lower acquisition cost by converting high-intent research directly to purchase.
- About 10–20% fewer service contacts through clearer comparisons, delivery promises, and order handoff.

Growth

- Expanded reach as third-party agents can discover and transact with the brand while honoring policy.
- Larger baskets through relevant bundles and substitutes (about 5–10% increase in order value).



Roles addressed

- Content marketing managers
- Analytics and data teams
- Marketing leadership
- Search engine optimization

How – Underlying foundation¹

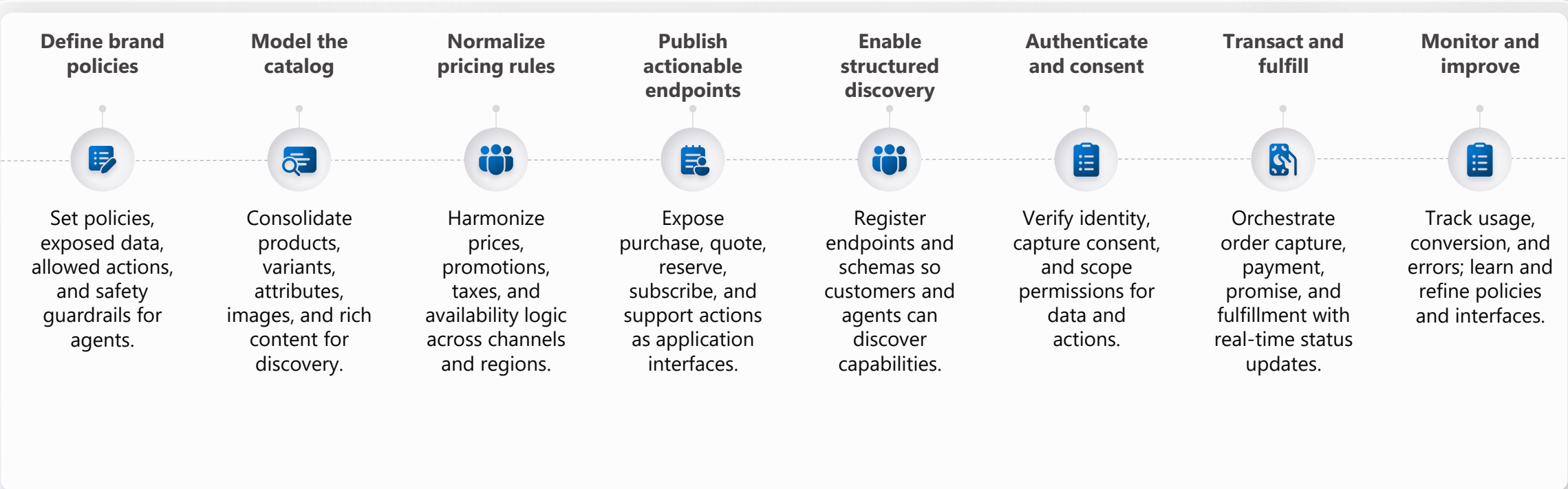
Data to integrate

- Product catalogs and specification feeds (including images, manuals, warranties).
- Pricing, promotions, taxes, delivery options, and return policies.
- Inventory by location, delivery calendars, pickup options, and shipping service levels.

Technical patterns

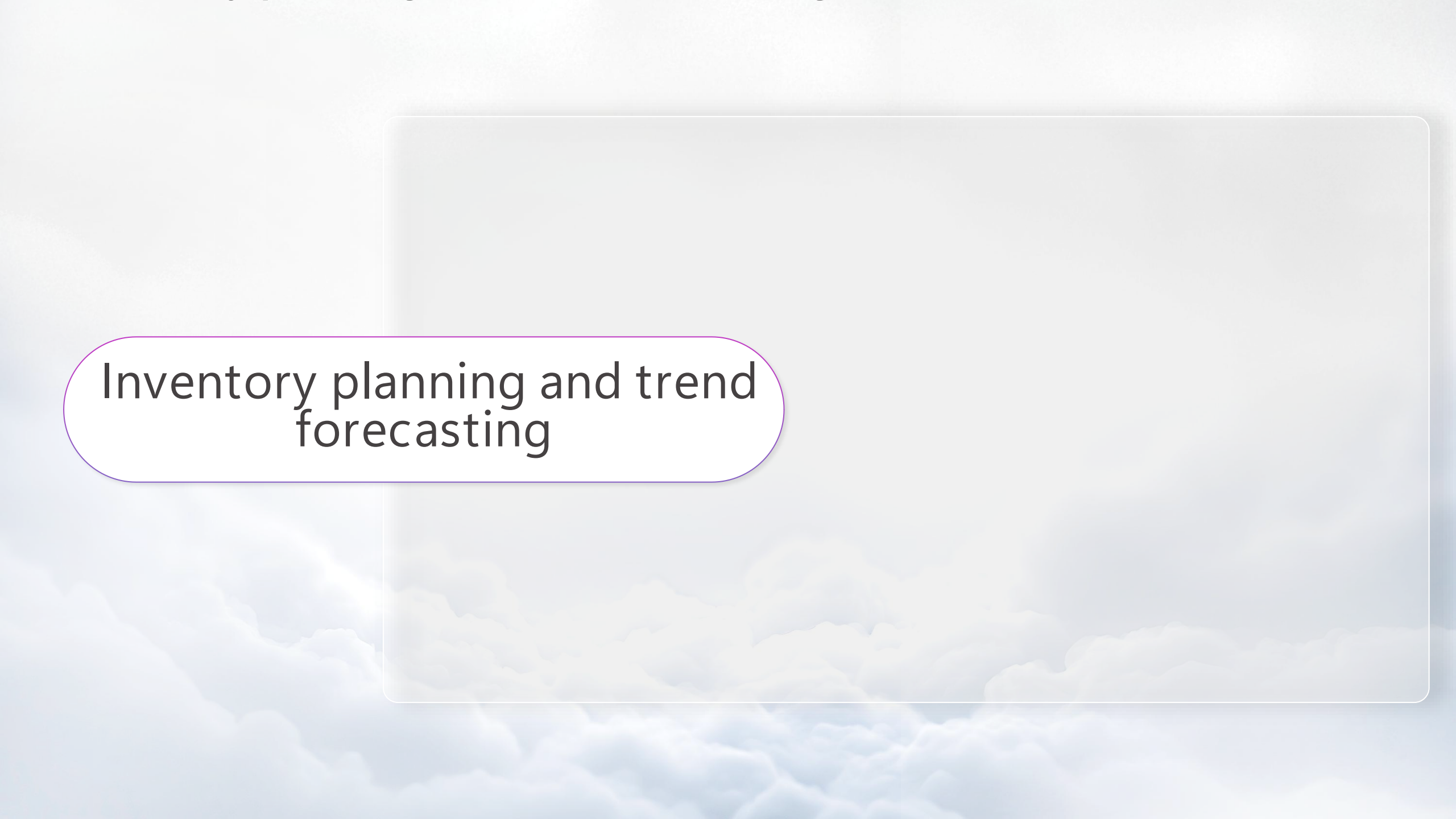
- Federated search and aggregation with entity resolution and specification normalization across merchants.
- Secure action and data exposure through documented application programming interfaces, including schema-based product data, consent, and policy enforcement for checkout and order relay.

3. What does the commerce process look like?



Key Business Capabilities required for this process

Catalog data management	Pricing and promotion engine	Availability and promise service	Policy and guardrail engine	Action schema definition
Application interface management	Identity and access control	Consent and privacy controls	Order and payment orchestration	Observability and performance analytics



Inventory planning and trend forecasting

4. Inventory planning and trend forecasting

Demand sensing fuses real-time signals from point-of-sale systems, online commerce, promotions, and external drivers like weather to predict near-term demand by item and location. Accurate, granular forecasts reduce stockouts and excess inventory, align production and replenishment, and improve service levels while lowering working capital and waste.

What - Process overview



Current challenges

- Fragmented data across channels; slow batch updates miss rapid demand shifts.
- Forecasts averaged at regional or weekly level hide store-item volatility.
- Manual overrides and disconnected planning tools lead to bias and rework.
- Limited incorporation of external factors (weather, events, competitor moves).
- Bullwhip effects across suppliers from poor signal sharing and long lead times.



Frontier solution

Process description

- Continuously ingest transactional and behavioral signals from stores, online, and partners
- Cleanse, align, and enrich data at item–location–day granularity with event and weather features
- Generate short-term trend forecasts and medium-term plans; detect anomalies and shifts
- Recommend inventory targets and replenishment orders; simulate scenarios and constraints
- Publish decisions to planning, warehouse, and transportation systems; track accuracy and drift

Solution description

Agents collaborate via events, share context, and hand off to humans for final sign-off:

- **Signal Ingestion Agent** aggregates point-of-sale, online orders, returns, and promotion calendars; validates quality and freshness.
- **Feature & Causality Agent** builds event, price, and weather features; tests impact and selects drivers.
- **Forecasting Agent** produces item–location forecasts (short and medium term) with uncertainty ranges; flags anomalies.
- **Replenishment & Allocation Agent** converts forecasts into targets and orders by store or channel under constraints.
- **Planner Copilot** explains drivers, suggests overrides with guardrails, and coordinates hand-offs to production and logistics.

Why – Impact¹



Business Outcomes²

Cost savings:

- 10–20% reduction in forecast error; 20–40% fewer stockouts and expedites.
- 5–15% lower inventory and carrying costs; 10–25% less waste and obsolescence

Growth

- 1–3 percentage-point increase in service level and on-shelf availability.
- 2–5% uplift in revenue from better availability during promotions and peaks.



Roles addressed

- Supply and demand planning leaders.
- Merchandising and replenishment managers.

How – Underlying foundation¹

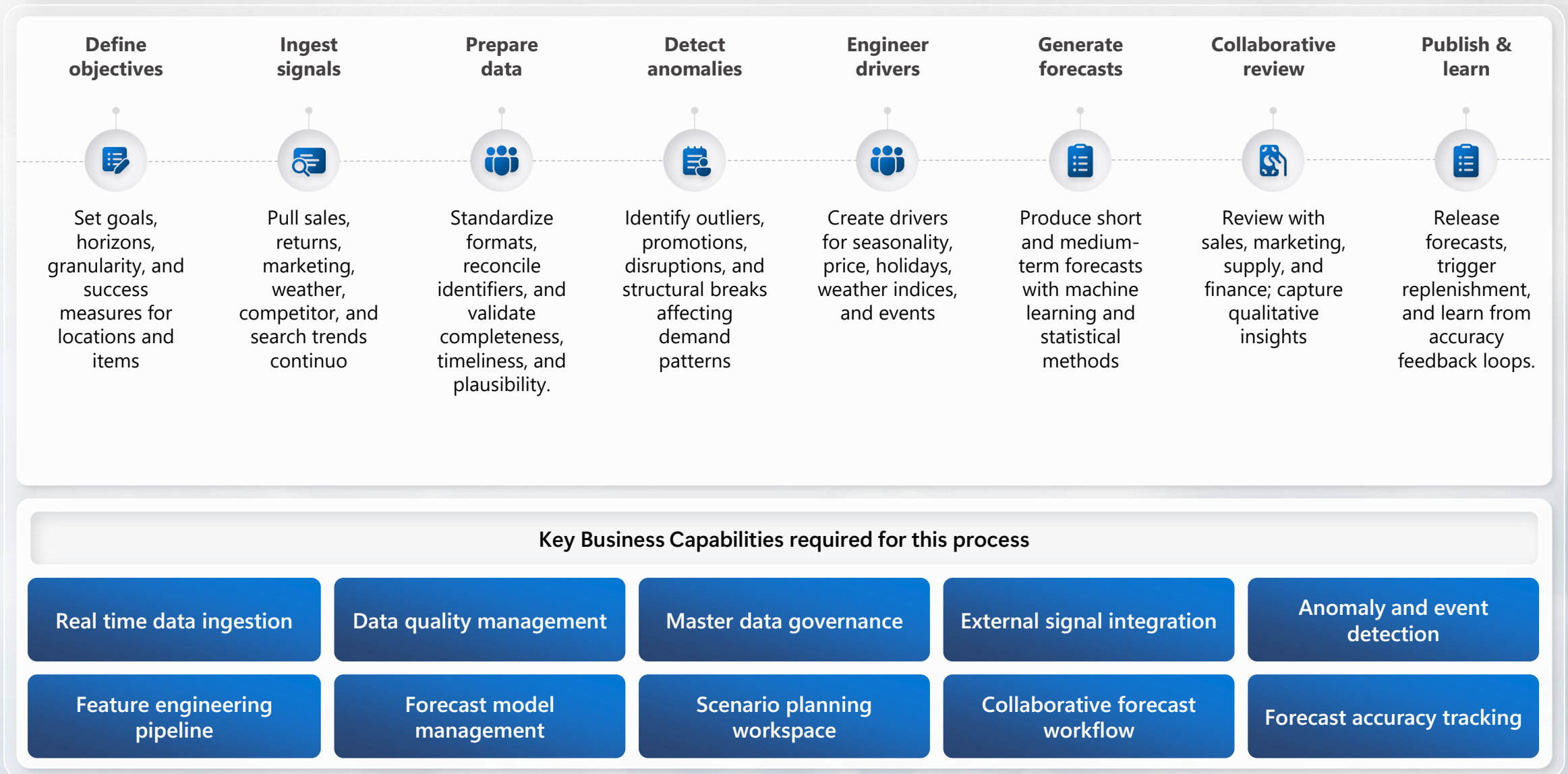
Data to integrate

- Sales and orders by item and location (store and online) with returns and cancellations.
- Promotion, price, catalog, and inventory positions with lead times and constraints.
- External signals: weather, local events, holidays, and competitor activity where available.

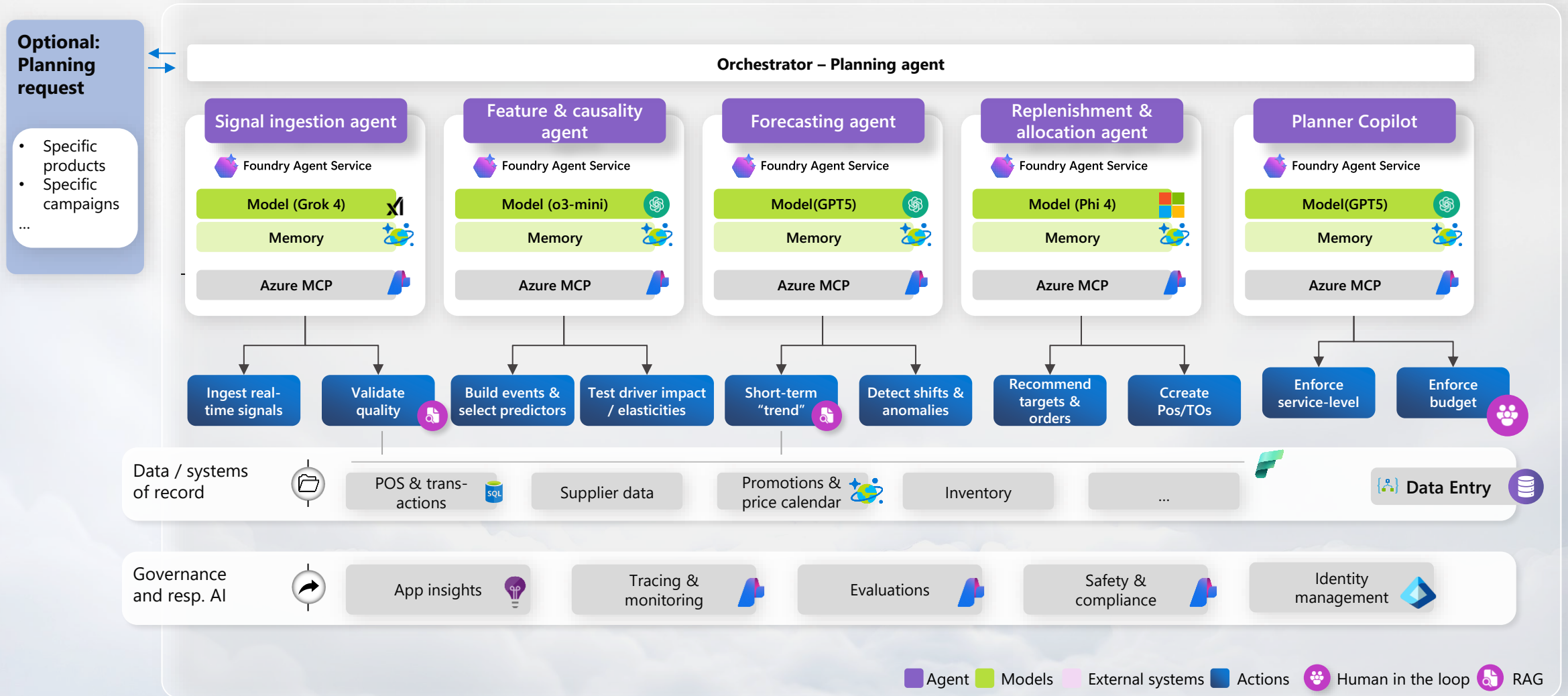
Technical patterns

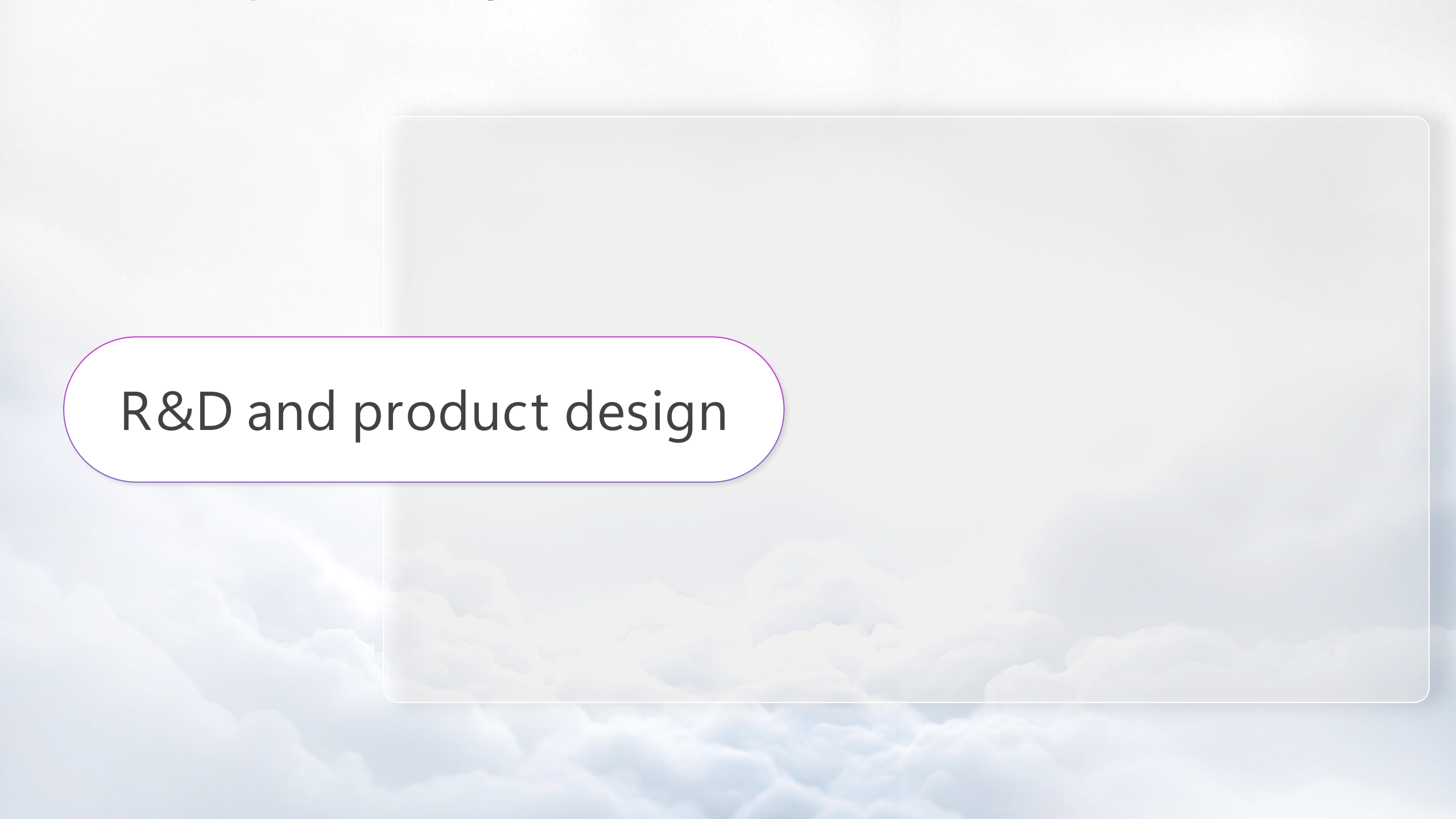
- Event-driven data pipelines with feature store and retrieval-augmented forecasting that supports tool use into planning systems.
- Human-in-the-loop governance with traceable decisions, model monitoring, and drift detection.

4. Inventory planning & trend forecasting process



4. Example agentic inventory planning and trend forecasting





R&D and product design

5. R&D and product design

Agentic product innovation continuously converts trends, customer signals, and regulatory changes into clear portfolio actions. The goal is to surface gaps, draft concept briefs and targeted design tweaks, and run controlled experiments so teams compress time-to-market, reduce launch risk, and raise the probability of commercial success.

What - Process overview



Current challenges

- Trend research is manual and scattered; weak signal detection leads to late insights.
- Customer feedback is noisy; teams struggle to separate preferences from complaints.
- Concept briefs take weeks; iteration cycles depend on meetings and handoffs.
- Experiments are inconsistent across channels; learning is hard to compare or reuse.
- Regulatory and safety constraints are applied late, causing costly rework.



Frontier solution

Process description

- Continuously ingest trend sources, customer input, sales patterns, and regulatory updates.
- Cluster needs and opportunities; score by demand, feasibility, and strategic fit.
- Auto-draft concept briefs with value propositions, target segments, and early designs.
- Propose packaging or formula tweaks; generate design variants tied to measurable goals.
- Orchestrate controlled experiments across sites or channels; collect outcomes and decide go, refine, or stop.
- Maintain a living portfolio view that tracks assumptions, evidence, and next actions.

Solution description

Agents collaborate via events, share context, and hand off to humans for final sign-off:

- **Signal ingestion agent** fuses social, search, sales, and help-desk signals; highlights emerging needs and white spaces
- **Voice-of-Customer Synthesis Agent** summarizes reviews, and returns; extracts desired outcomes and pain points
- **Concept Brief Generator** produces drafts with problem statement, persona, constraints, and early bill-of-materials or design notes
- **Compliance & Feasibility Agent** checks ingredients, materials, labeling, and safety rules; flags risks and alternatives.
- **Experiment Orchestrator** designs-controlled tests, defines success metrics, launches variants, and compiles decisions and learnings

Why – Impact¹



Business Outcomes²

Cost savings:

- 20–30% faster concept-to-brief cycle through automated drafting and review.
- 15–25% fewer late-stage reworks by applying feasibility and compliance checks up front.

Growth

- Higher launch success rate (expected +5–10 percentage points) from evidence-based gating.
- 2–5% revenue uplift from faster response to trends and customer needs.



Roles addressed

- Product management and innovation leaders.
- Research and design directors.

How – Underlying foundation¹

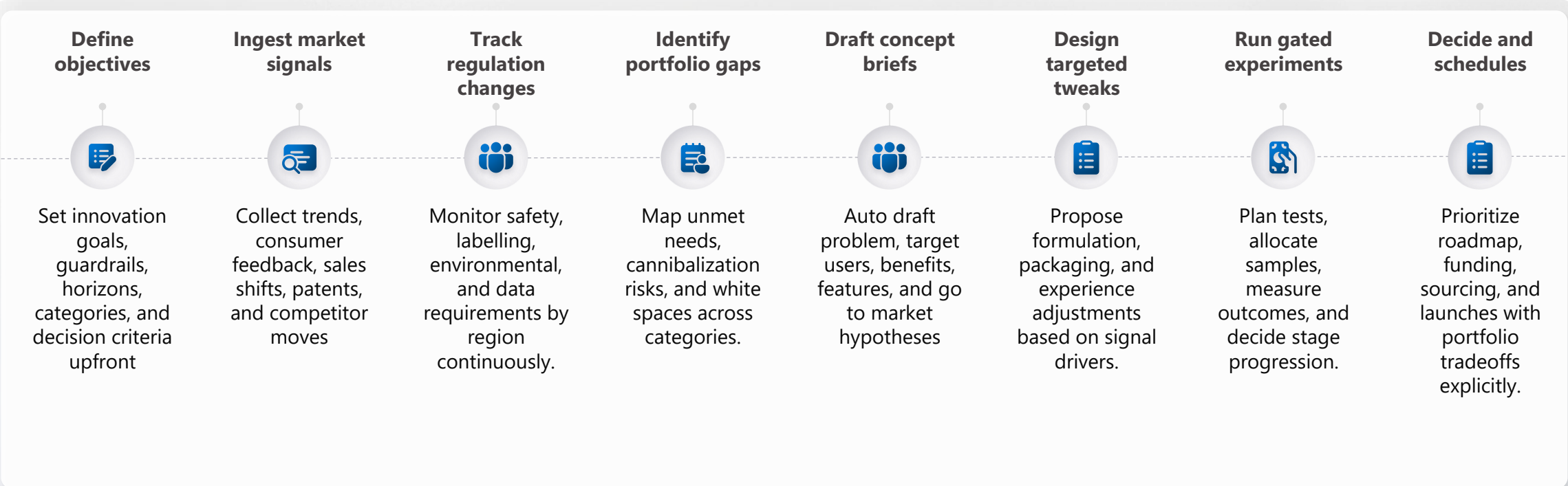
Data to integrate

- Customer signals: reviews, support conversations, and satisfaction surveys.
- Market and trend sources: search trends, social mentions, competitor listings, and analyst reports
- Regulatory and safety rules, materials and ingredient libraries, and mfg. constraints

Technical patterns

- Multimodal retrieval and synthesis with grounded citations over product, market, and policy data.
- Experiment orchestration with event-driven workflows, telemetry, and human-in-the-loop governance.

5. What does the dynamic inventory optimization process look like?



Key Business Capabilities required for this process				
Trend and signal ingestion	Consumer feedback analytics	Sales and demand sensing	Regulatory intelligence tracking	Portfolio gap analysis
Concept brief generation	Design and specification management	Experiment design and testing	Financial impact modeling	Portfolio governance workflow



Thank you